

$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}, \quad x > 0, y > 0$$

Final Graphical Interface

Beta Function Calculator

Compute B(x, y) for finite positive real inputs.

Inputs

x (Alt+X)2y (Alt+Y)3

CalculateSwap x and yClear

Result and status

B(2, 3) = 0.0833333333333

Calculation completed.

Calculation history

x	y	B(x, y)
2	3	0.0833333333333

Enter: calculate Esc: clear Alt+X / Alt+Y: focus input

The live interface demonstrates a known value, visible calculation status, and persistent history. Native Tkinter widgets preserve familiar focus and tab behavior.

Accessibility and Recovery

Keyboard access
Enter, Esc, Alt+X, Alt+Y

Recognition over recall
visible labels and actions

Helpful recovery
field-specific text messages

Status is not color-only
words identify success or error

The GUI aims to be accessible through readable labels, visible focus, redundant text feedback, keyboard operation, and recoverable validation. These choices align the interface with the user’s task model (Kamthan, 2026c).

Exception policy: expected input and numeric-range errors receive specific messages; unexpected arithmetic failures are not hidden behind a generic message and remain available for diagnosis.

D3 conclusion: the release improves numerical stability, accessibility, and software quality through a stable log-beta implementation, explicit range behavior, and repeatable verification.

Problem 7: Style and Static Analysis

```
wei Huang@Weis-MacBook-Pro D3_F6_Beta_Function % clear
.venv/bin/flake8 beta_core.py beta_function_gui.py test_beta_core.py debug_beta.py verify_accuracy.py && echo "Flake8 passed: no violations found."
Flake8 passed: no violations found.
wei Huang@Weis-MacBook-Pro D3_F6_Beta_Function %

wei Huang@Weis-MacBook-Pro D3_F6_Beta_Function % clear
.venv/bin/pylint beta_core.py beta_function_gui.py test_beta_core.py debug_beta.py verify_accuracy.py

-----
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)
```

Flake8 reports no PEP 8 style violations; Pylint rates the final source **10.00/10**. Both outputs were generated from version 1.0.0.

Problem 7: Debugger Evidence

```
wei Huang@Weis-MacBook-Pro D3_F6_Beta_Function % clear
python3 -m pdb debug_beta.py
> /Users/wei Huang/Documents/SOEN 6011/D3_F6_Beta_Function/debug_beta.py(1)<module>()
-> """Small driver used only to demonstrate pdb on the Beta calculation."""
(Pdb) break beta_core.beta_function
Breakpoint 1 at /Users/wei Huang/Documents/SOEN 6011/D3_F6_Beta_Function/beta_core.py:256
(Pdb) continue
> /Users/wei Huang/Documents/SOEN 6011/D3_F6_Beta_Function/beta_core.py(258)beta_function()
-> if not is_finite(x_value):
(Pdb) p (x_value, y_value)
(2.0, 3.0)
(Pdb) next
> /Users/wei Huang/Documents/SOEN 6011/D3_F6_Beta_Function/beta_core.py(260)beta_function()
-> if not is_finite(y_value):
(Pdb) next
> /Users/wei Huang/Documents/SOEN 6011/D3_F6_Beta_Function/beta_core.py(262)beta_function()
-> if x_value <= 0.0:
(Pdb) next
> /Users/wei Huang/Documents/SOEN 6011/D3_F6_Beta_Function/beta_core.py(264)beta_function()
-> if y_value <= 0.0:
(Pdb) continue
B(2, 3) = 0.0833333333333333
The program finished and will be restarted
> /Users/wei Huang/Documents/SOEN 6011/D3_F6_Beta_Function/debug_beta.py(1)<module>()
-> """Small driver used only to demonstrate pdb on the Beta calculation."""
(Pdb) quit
wei Huang@Weis-MacBook-Pro D3_F6_Beta_Function %
```

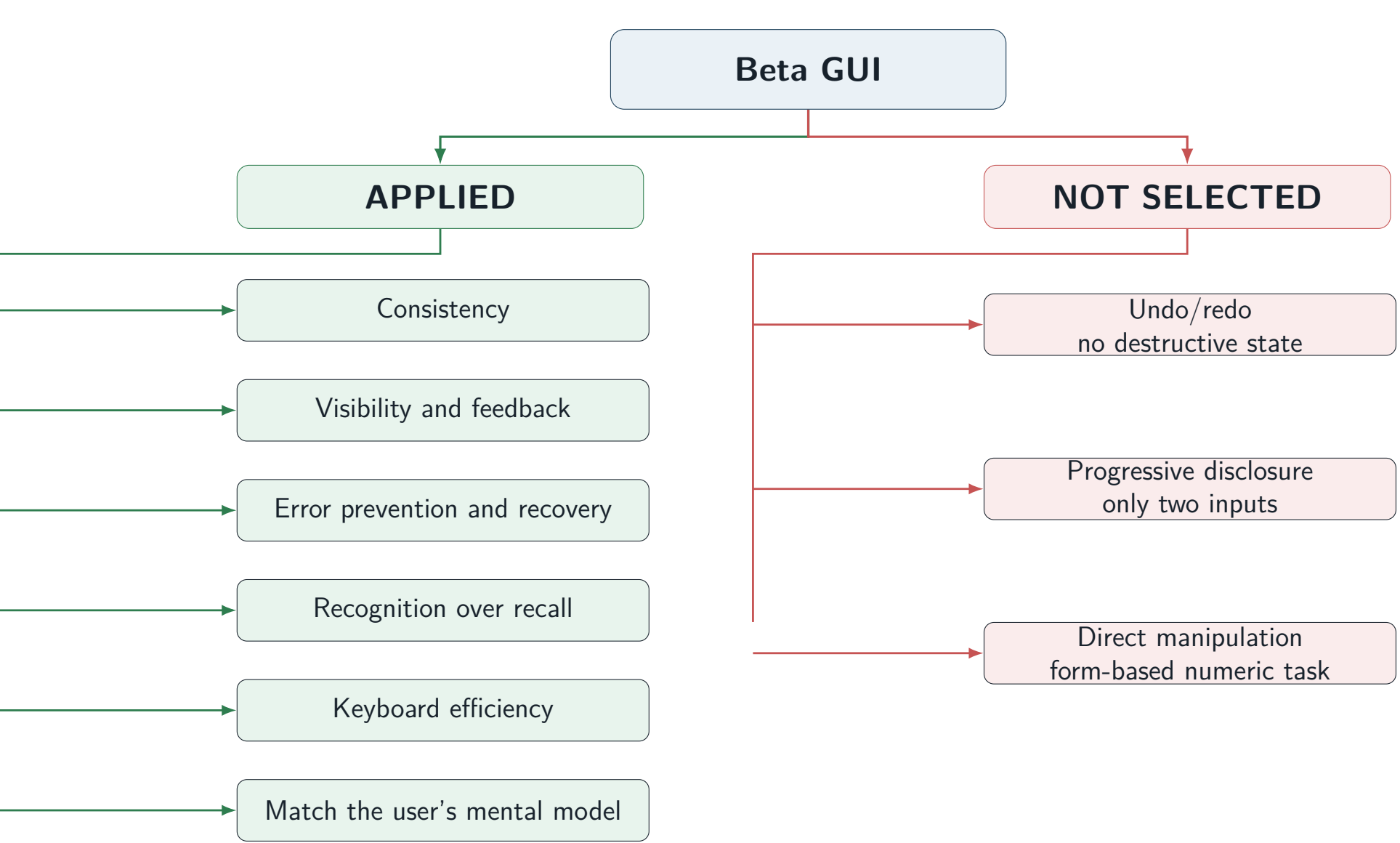
A reproducible pdb session stops inside beta_function, inspects (x, y) = (2, 3), steps through validation, and confirms the known result 1/12.

Verified Numerical Contract

```
wei Huang@Weis-MacBook-Pro D3_F6_Beta_Function % clear
.venv/bin/python verify_accuracy.py
External verification only - mpmath, 100 decimal digits
Seed: 6011; generated cases: 500
Inputs: log-uniform from 1e-12 through 1e12
Normal representable results checked: 414
Range-limited reference results excluded: 86
Failures above 1e-10 relative error: 0
Maximum relative error: 1.223e-11
Worst normal-result input: (131117.64197605842, 81.32262207870075)
Large regression B(1e100, 1) relative error: 5.419e-13
Roundable boundary exp(-745): 4.9406564584124654e-324
wei Huang@Weis-MacBook-Pro D3_F6_Beta_Function %
```

Contract: finite positive inputs produce a representable binary64 result or a specific range error. Twelve displayed digits are formatting, not a universal accuracy claim; subnormal values follow binary64 spacing.

UIDP Decision Mind Map



The mind map records both selected and rejected principles instead of treating every principle as mandatory (Kamthan, 2026d).

Problem 8: PyUnit Test Suite

```
wei Huang@Weis-MacBook-Pro D3_F6_Beta_Function % clear
python3 -m unittest -v
test_half_half_equals_pi (test_beta_core.BetaFunctionTests.test_half_half_equals_pi)
B(0.5, 0.5) equals pi. ... ok
test_known_value_one (test_beta_core.BetaFunctionTests.test_known_value_one)
B(1, 1) equals one. ... ok
test_known_value_two_three (test_beta_core.BetaFunctionTests.test_known_value_two_three)
B(2, 3) equals one twelfth. ... ok
test_large_asymmetric_identity (test_beta_core.BetaFunctionTests.test_large_asymmetric_identity)
B(x, 1) remains accurate when x is extremely large. ... ok
test_near_maximum_asymmetric_identity (test_beta_core.BetaFunctionTests.test_near_maximum_asymmetric_identity)
A near-maximum finite x does not trigger log-gamma cancellation. ... ok
test_nonfinite_input_is_rejected (test_beta_core.BetaFunctionTests.test_nonfinite_input_is_rejected)
Infinite input is rejected before calculation. ... ok
test_nonnumeric_input_is_rejected (test_beta_core.BetaFunctionTests.test_nonnumeric_input_is_rejected)
The GUI parser reports a useful nonnumeric-input error. ... ok
test_roundable_subnormal_exponential (test_beta_core.BetaFunctionTests.test_roundable_subnormal_exponential)
A finite result near the subnormal boundary is not rejected. ... ok
test_scientific_notation_parser (test_beta_core.BetaFunctionTests.test_scientific_notation_parser)
The GUI parser accepts scientific notation. ... ok
test_symmetry (test_beta_core.BetaFunctionTests.test_symmetry)
Swapping x and y does not change the result. ... ok
test_underflow_is_reported (test_beta_core.BetaFunctionTests.test_underflow_is_reported)
An unrepresentable positive result is not returned as zero. ... ok
test_unexpected_arithmetic_error_is_not_hidden (test_beta_core.BetaFunctionTests.test_unexpected_arithmetic_error_is_not_hidden)
Unexpected arithmetic failures remain visible for diagnosis. ... ok
test_unroundable_exponential_is_rejected (test_beta_core.BetaFunctionTests.test_unroundable_exponential_is_rejected)
An exponential below the rounding boundary reports underflow. ... ok
test_zero_is_rejected (test_beta_core.BetaFunctionTests.test_zero_is_rejected)
Zero is rejected because it is outside the Beta domain. ... ok

Ran 14 tests in 0.002s

OK
wei Huang@Weis-MacBook-Pro D3_F6_Beta_Function %
```

14/14 tests pass. Coverage includes trusted values, symmetry, extreme asymmetric inputs, the roundable subnormal boundary, scientific notation, invalid input, underflow, and propagation of unexpected arithmetic errors. The tests exercise the final numerical core and parser.

Generative AI and CASTROFF. Tool: OpenAI ChatGPT/Codex. Problem 7 prompt: “Review the stable log-beta code for PEP 8, Flake8, pdb, Pylint, SemVer, UIDP, accessibility, and honest range behavior.” Problem 8 prompt: “Propose PyUnit regressions for large inputs, subnormal rounding, invalid input, and unexpected exceptions.” Context: SOEN 6011 D3; Audience: TA/instructor; Style: concise; Task: review; Role: student engineer; Output: suggestions; Format: poster checklist; Feedback: accept only after execution. Suggestions were revised and verified; unverified claims were rejected.

References. Kamthan, P. (2026a), *SOEN 6011 Project Description*. Kamthan, P. (2026b), *Building, Versioning, and Releasing Software*. Kamthan, P. (2026c), *Introduction to Mental Models*. Kamthan, P. (2026d), *Brainstorming and Mind Mapping*. Van Rossum, G., Warsaw, B., and Coghlan, N. (2001), *PEP 8 – Style Guide for Python Code*. Python Software Foundation (2026), *unittest and pdb Documentation*.

Repository: github.com/HW925/SOEN-6011-D3-Beta-Function